NOT MEASUREMENT SENSITIVE

MIL-I-15665H
29 April 1994
SUPERSEDING
MIL-I-15665G
15 March 1988

MILITARY SPECIFICATION

INSIGNIA; BRANCH OF SERVICE, HAT, GARRISON CAP, AND SERVICE CAP; ENLISTED PERSONNEL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers branch of service insignia, hat insignia, garrison cap insignia, and service cap insignia used by enlisted personnel of the Armed Forces (see 6.1).

1.2 Classification. The insignia covered by this specification shall be as specified on the applicable military specification sheet (see 2.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Shipping, Fiberboard
PPP-T-97 - Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, U.S. Army, 9325 Gunston Road, Room S112, Fort Belvoir, VA 22060-5579, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455

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MILITARY

MIL-F-495 - Finish, Chemical, Black for Copper Base Alloy

STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods
FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

SPECIFICATION SHEETS

(See supplement 1 for list of associated specifications.)

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

THE INSTITUTE OF HERALDRY (TIOH)

4-4-89 - Insignia of Branch, Enlisted Personnel, Assembly and Details
B-13-12 - Attaching Devices, Heraldic, Prong and Clutch Type
B-2 (series) - Branch of Service, Enlisted Personnel

STANDARD COLOR CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD HARD ENAMEL COLOR CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD EPOXY COLOR CHIP SET

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD METAL FINISH CHIP SET

(The standard color chip(s) for color or finish may be obtained from the procuring activity for Government procurements.)

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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(Figure 1 and Figure 2 are miniature copies of Institute of Heraldry drawings and are for information only.)

2.2 Non-Government publications. The following documents form a part of this document to the extend specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-86 - Zinc-Alloy Die Casting
ASTM-B-487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of
ASTM-D-2240 - Rubber Property - Durometer Hardness
ASTM-D-3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Recovered material shall be used to the maximum extent possible.

3.3.1 Zinc aluminum alloy. The alloy used for die casting shall conform to alloy AG 40a or AC 41a of ASTM-B-86.

3.3.2 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregations and foreign inclusions that will not be removed in later processing. When tested as specified in 4.6.1, the chemical composition of the copper base alloy shall be a specified in Table I.

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TABLE I. CHEMICAL COMPOSITION

Alloy	Copper	Nickel 1/	Lead (Max)	Iron (Max)	Manganese (Max)	Zinc	Others (Max)
Yellow Brass	64.0 - 68.5	--	0.15	0.05	--	Remainder	0.15
Red Brass	84.0 - 86.0	--	0.05	0.05	--	Remainder	0.15
Gilding Metal	94.0 - 96.0	--	0.03	0.05	--	Remainder	0.13
Free cutting Brass	60.0 - 63.0	--	2.50-3.70	--	--	Remainder	--
Bronze	89.0 - 91.0	--	0.05	0.05	--	Remainder	0.13
Nickel Silver	63.0 - 65.5 2/	16.5 - 19.5 3/	0.10	0.25	0.50	Remainder	--
Low Brass	78.5 - 81.5	--	0.05	0.05	--	Remainder	0.15

1/ Cobalt counting as nickel.
2/ Copper content for the nickel silver pin only may be 53.5 to 56.5%.
3/ When nickel silver is used as a base material, the nickel content shall be not less than 18%.

3.3.3. 24 karat gold for plating. Gold for plating shall be 24 karat.

3.3.4. 22 karat gold for plating. Insignia may be manufactured using 22 karat gold plate with nickel underplating when specified on the applicable specification sheet for commercial sale only.

3.3.5. Nickel for plating. Nickel for plating shall be the nickel plating normally used in commercial practice.

3.3.6. Silver for plating. Silver for plating shall be not less than 99.5 percent pure silver.

3.3.7 Solder.

3.3.7.1 Hard Solder. Hard solder shall be a silver alloy solder having a melting point of not less than 1075 degrees Fahrenheit (561.7 degrees Celsius).

3.3.7.2 Soft Solder. Soft solder shall be a lead-tin alloy having a melting point of not less than 375 degrees Fahrenheit (190.58 degrees Celsius).

3.3.8 Enamel.

3.3.8.1 Hard enamel. Hard enamel shall be a glass, vitreous type enamel fused with metallic oxides to produce the required color, opacity or translucence, and shade.

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3.3.8.2 Epoxy resin (stoning epoxy). Epoxy resin shall be a stoning epoxy, have a shore D hardness of 85 and incorporate an ultraviolet inhibitor.

3.3.8.3 Baking enamel. The enamel for subdued insignia shall be a flat synthetic, baking enamel pigmented to the shade of black matching the standards established in FED-STD-595 for Color Chip number 37038.

3.3.9 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of a pigmented lacquer shall not be permitted.

3.4 Design. The embossed design of the insignia shall be an exact duplicate of the design on the Government loaned hubs (see 3.7), from which the contractor's working dies shall be extracted. The contractor's working die shall be tooled and polished to remove any dents, nicks, scratches or other imperfections.

3.5 Construction. Army branch of service insignia shall be constructed in accordance with details and dimensions shown on Drawing 4-4-89 (Figure 1).

3.5.1 Stamping, trimming and piercing. The stamping shall have a well defined die struck edge. The insignia shall be trimmed and, when applicable, pierced to the die struck edge. All edges shall be clean, smooth and free from burrs, drag, step and rough edges. The stamping, piercing and trimming operations shall not damage or distort the design or alter the shape of the item.

3.5.2 Die casting. When die cast method is applicable, the insignia shall be free from porosity, blow holes or other casting defects. Before finishing, all flash lines shall be removed and the edges of the insignia shall be smooth. Insignia that is die cast from zinc aluminum specified in 3.3.1 shall be copper plated. When tested as specified in 4.6.3.2, the plating shall be a minimum of 0.0003 inch thick (0.007 cm).

3.5.3 Soldering and electronic fusion. Unless otherwise specified on the applicable specification, all soldering shall be accomplished using hard solder specified in 3.3.7.1 or by electronic fusion. Joints shall be clean, smooth, strong and free from burned or reduced areas. There shall be no excess solder and all excess flux shall be removed. When hard solder is used, the soldered part and prong shall not separate at the joint when tested as specified in 4.6.4.1. When fused joints are used, the prong shall not separate at the joint when tested as specified in 4.6.5.

3.5.4 Forcers. When forcers are permitted to bring up the design, the back of the insignia shall have no sharp or rough edges. The impression of the forcer shall be symmetrical with the embossed design of the insignia. Sufficient metal shall be left near any trimmed area to assure clean smooth trimming without distortion. Unless otherwise specified on the applicable specification sheet, mock forcers shall not be permitted. When mock forcers are permitted, the forcer shall be used only on the areas specified.

3.5.5 Attaching device. Attaching devices shall be as specified on the applicable specification sheet except for Army branch of service and collar insignia which shall have a back plate with prong and clutch type attaching device as specified on Drawing 4-4-89 (Figure 1). The back plate shall be fabricated of any type of 1/4 hard brass. As an alternative, the back plate and prongs may be die cast as specified in 3.5.2; however, the back plate shall be brass plated after

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copper plating. When prongs are tested in accordance with 4.6.4.1, no evidence of breaking or cracking shall occur.

3.5.5.1 Prong (registering). Prongs shall be made from nickel silver specified in 3.3.2. The prongs shall be of the dimensions shown on the applicable specification sheet. They shall be positioned and attached as required by the applicable specification sheet. Registering prongs shall be an integral part of the insignia when die cast zinc aluminum is used (see 3.3.1).

3.5.5.2 Prong (integral). When prongs are die cast as integral part of the insignia and tested in accordance with 4.6.3.1, no evidence of breaking or cracking shall occur.

3.5.5.3 Screw post type and thumb nut. Screw post and thumb nut attaching devices shall be made from free cutting brass specified in 3.3.2 and shall conform to the details and dimensions shown on the applicable specification sheet. The screw post shall be hard soldered or electronically fused to the back of the insignia in the location shown. When tested as specified in 4.6.9.2, thumb nuts shall not bind on the post's threads. Screw post shall be an integral part of insignia when die cast zinc aluminum is used (see 3.3.1).

3.5.5.4 Prong and clutch type. Unless otherwise specified, prongs shall be made from nickel silver specified in 3.3.2. Unless otherwise specified on the applicable specification sheet, the prongs as shown on Drawing 4-4-89 (Figure 1) shall be 5/16 inch $\pm$ 1/64 inch (0.79 cm $\pm$ 0.04 cm) long. The prongs shall be driven, swaged, hard soldered or electronically fused in the locations specified in Drawing 4-4-89 (Figure 1) or the applicable specification sheet. Prongs for Army branch of service insignia shall be made from hard drawn brass. Clutches as shown on Drawing B-13-12 (Figure 2) shall be made from any type brass. When tested as specified in 4.6.9.1, it shall not be possible to remove the clutch from the prong without first depressing the release wings. When prongs are die cast as an integral part of the insignia and tested in accordance with 4.6.3.1, no evidence of breaking or cracking shall occur.

3.6 Finish. Unless otherwise specified, the front and edges (surfaces visible from the front) shall be finished as specified on the applicable specification sheet and shall match the TIOH standard enamel color or metal finish chips as required (see 2.1.2). Where plating or oxidizing is specified, the entire item (front, back and edges), except attaching devices, shall be plated or oxidized. Unless otherwise specified, the backs of all oxidized items shall be clean and uniformly oxidized. Army branch of service insignia and collar insignia may be gold plated for commercial sale. In addition to the finish specified, all insignia, except Marine Corps black enameled insignia and gold plated insignia, shall be thoroughly coated (front, back and edges) with clear lacquer specified in 3.3.9. Army branch of service and collar insignia need only be lacquered on the front surface and edges. At the option of the manufacturer, any gold plated insignia may be lacquered. Marine Corps black colored insignia shall be given a black chemical finish (front, back and edges) conforming to MIL-F-495. The face and edges of the insignia shall then be coated with flat dead black enamel specified in 3.3.8.3.

3.6.1 Enameling. Unless otherwise specified on the applicable specification sheet, all enameling shall be accomplished using hard enamel. Epoxy resin may be used in lieu of hard enamel at the option of the manufacturer. Enamel shall be applied so as to be uniform in color, free from bubbles, foreign inclusions,

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cracking, crazing, or other defects which might affect appearance. There shall be no overrunning of enamel.

3.6.1.1 Colors. Enamel or epoxy colors shall match the shades of the Institute of Heraldry color chips specified on the applicable specification sheet (see 2.1.2).

3.6.1.2 Hard enameling. Hard enamel specified in 3.3.8.1 shall be charged, fired and stoned level with the dikes. No design elements shall be removed during the stoning process. There shall be no noticeable burn spots as a result of the firing. The hard enamel shall then be lapped, felted, or wheel polished to produce a glass like finish. The finish shall be adherent and free from bubbles, pits, foreign inclusions, cracking, crazing, burned edges or darkened enamel.

3.6.1.3 Epoxy Resin. Epoxy resin specified in 3.3.8.2 shall be applied and stoned level with the dikes. No design elements shall be removed during the stoning process. The epoxy resin shall then be lapped, felted, or wheel polished to produce a glass like finish. When tested as specified in 4.6.6, the epoxy shall have a shore D hardness of not less than 85.

3.6.1.4 Baking enamel. When insignia is to be subdued, the face and edges shall be coated with enamel specified in 3.3.8.3. Prior to enameling, the entire insignia shall be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.6.8, the baked enamel shall not flake or chip. When tested as specified in 4.6.7, only a slight amount of color shall be permitted to transfer to the cotton.

3.6.2 Plating. All plating shall be by electroplating methods. The plating shall be continuous and unbroken over the entire plated surface. There shall be no cut-through, shade, peeled, or blistered plating. Plating of attaching devices will not be required.

3.6.2.1 24 karat gold plating. The gold plating shall be accomplished using gold specified in 3.3.3. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating for gold shall not be permitted. When tested as specified in 4.6.2, no visible chemical reaction such as evolution of gases shall appear.

3.6.2.2 22 karat gold plating. The gold plating shall be accomplished using gold specified in 3.3.4. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating for gold shall be permitted. When tested as specified in 4.6.2, no visible chemical reaction such as evolution of gases shall appear.

3.6.2.3 Nickel plating. Nickel for plating shall be the nickel plating normally used in commercial practice.

3.6.2.4 Nickel underplating. When nickel underplating is specified, the underplating shall be not less than 0.00025 inch (0.000635 cm) thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.6.2.5 Silver plating. When silver plating is specified, silver specified in 3.3.6 shall be used. After finishing, the silver plating shall be not less than 0.0003 inch (0.0008 cm) thick. Items made of brass shall be silver thickness tested

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on their obverse side. In case of a dispute, testing shall be conducted in accordance with ASTM-B-487.

3.6.3 Lacquering. All insignia, except Marine Corps black enameled insignia and gold plated insignia, shall be lacquered using lacquer specified in 3.3.9 as applicable (see 3.6 and appropriate specification sheet). The dry lacquer film shall be continuous, level, adherent and free from lint, dust or other foreign matter. When tested as specified in 4.6.10.1, there shall be no change in appearance of the lacquered surface except for a slight yellowing of the highlights. When tested as specified in 4.6.10.2, the tissue paper shall not adhere to the clear lacquered surface.

3.7 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order (see 6.4).

3.8 Marking for identification. Except for the Air Force service cap insignia, the Air Force branch of service insignia, Marine Corps branch of service insignia and female hat insignia, the contractor shall stamp his trade or other identifying mark, the letters "GI" (GI for government procurement only) legibly and inconspicuously on the back of each item. When practical, the items will be stamped in the forced areas. Marking shall be placed on the attaching bar for Army branch of service insignia and on the washer or thumb nut for the Air Force service cap insignia, Marine Corps branch of service insignia, and Marine Corps service cap insignia. Female hat insignia and Air Force branch of service insignia need not be marked.

3.9 Workmanship. The finished item shall be clean, well made, and shall conform to the acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.4 First article inspection. Inspection and testing of the first article (see 3.2) shall be made of a completely finished item for all provisions of this specification applicable to the end product examination and tests.

4.5 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated (see 6.3).

4.5.1 Inspection of components.

4.5.1.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed in Table II for the test characteristics shown therein.

Table II. TESTING OF COMPONENTS

Component and lot expressed in terms of	Characteristic	Rqm't Para	Test Method	Rqm't applicable to lot average	#determ per sample unit	Results Reported as	Sample Unit
Copper base alloy (1 lb) as applicable	Chemical composition	3.3.2	4.6.1	X	2 each element	Nearest 0.1% for each element	4 oz material each alloy or 10 prongs
Gold used for plating (1 oz.)	Karat	3.3.3 3.3.4	Standard Comm'l	X	2 composite	Nearest 0.1%	1/2 oz.
Silver for plating	Fineness	3.3.6	Standard Comm'l	X	2 composite	Nearest 0.1%	1 oz.

4.5.1.1.1 Certification of compliance. Materials listed below may be accepted on the basis of the contractor's certificate of compliance for requirements specified in applicable paragraphs of this specification.

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COMPONENTS	CHARACTERISTICS	RQM'T PARA
Zinc aluminum alloy	Material identification	3.3.1
Copper base alloy	Material identification	3.3.2
24 karat gold plating	Material identification and karat	3.3.3
22 karat gold plating	Material identification and karat	3.3.4
Nickel for plating	Material identification	3.3.5
Silver for plating	Material identification and fineness	3.3.6
Hard solder	Material identification	3.3.7.1
Soft solder	Material identification	3.3.7.2
Hard enamel	Material identification	3.3.8.1
Epoxy resin	Material identification	3.3.8.2
Baking enamel	Material identification	3.3.8.3
Lacquer	Material identification	3.3.9
Brass for prongs (when applicable)	Material identification	3.3.2
Nickel silver for prong (when applicable)	Material identification	3.3.2
Brass for screw post & thumb nut	Material identification	3.3.2
Brass for washer (Marine Corps only)	Material identification	As specified on applicable specification sheet
Back Plate (Army branch of service insignia)	Material identification and temper	3.5.5

4.5.1.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplished of the following is in accordance with the specification requirements.

REQUIREMENT, OPERATION OR ASSEMBLY	CHARACTERISTICS	RQM'T PARA
Black chemical finish	Used as a base	3.6.1.4
Nickel plating	Not used as an underplating	3.6.2.1

4.5.2 Inspection of the end item.

4.5.2.1 Visual examination of insignia. Examination shall be made at a distance of approximately 16 to 18 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. The defects found during examination shall be classified in accordance with 4.5.2.1.1. The unit of product for the examinations in 4.5.2.1.1 shall be one completely finished insignia.

4.5.2.1.1 Examination of insignia for defects in finish, design, material, construction, workmanship, and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

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TABLE III. DEFECTS

EXAMINE	DEFECT	CLASSIFICATION Major	(*)	Minor
FRONT:				
Color and finish	Poor match to TIOH Standard Metal Finish Chip	x	-	-
	Not required finish	-	*	-
	Discoloration, such as spot, stain or speck	-	*	-
	Not clean	-	-	x
	Paired insignia not matching in color, shade and finish (when applicable)	-	*	-
	Any solder spatter clearly noticeable	-	*	-
	Oxidized finish has not been sufficiently relieved or has been excessively relieved	-	*	-
	Background not finished as specified	x	-	-
Quality of metal (unplated items)	Surface is pitted, porous, crystalline, open grained or otherwise impaired	x	-	-
Hard Enamel and Epoxy	Not enameled (when required)	x	-	-
	Wrong color or shade	x	-	-
	Cracked, chipped or crazed	-	*	-
	Not stoned smooth, i.e., is course, uneven or has drag marks	-	*	-
	Not level with dikes, i.e., contains high or low spots	-	*	-
	Hole or blister	-	*	-
	Laps, overruns or skips	-	*	-
	Not clean, i.e., blemish such as spot, stain, spatter or imbedded foreign matter	-	*	-
Lacquer	Lacquer coating hazy, cloudy or powdering	-	-	x
	Foreign matter imbedded in finish, clearly noticeable	-	-	x
	Lacquer not smooth, continuous or adherent, i.e., flaking, blistering, rippling, peeling or has run, sag, or drops	-	*	-
	Lacquer not set to touch	-	-	x
Design	Construction or design details obscured, altered or not conforming to government hub.	x	-	-
	Not dapped (when required)	x	-	-
	Any significant detail not clear, missing or obliterated	x	-	-
	Any significant detail marred or reduced	-	*	-
	Any warp, twist or distortion, producing irregular surface contour or outline	-	*	-
	Any detail struck over resulting in a double impression	-	*	-
	Any area not pierced (when required)	-	*	-
Plating	Not plated when required or plating is not			

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TABLE III. DEFECTS (Continued)

EXAMINE	DEFECT	CLASSIFICATION Major	(*)	Minor
FRONT:				
	type specified	x	-	-
	Any plated area cut through, not continuous, broken, peeling, blistered or otherwise impaired	-	*	-
	Foreign matter imbedded	-	*	-
Workmanship and construction	Surface or edge not clean, not smooth or free from burrs, roughness, drag, step or tool marks	-	*	-
	Metal marks on exposed surface such as nick, dent, dig, gouge or scratch	-	*	-
	Die-struck edge, rim or outline is not well defined when applicable	x	-	-
	Not trimmed to the die-struck edge, when applicable	-	*	-
	Pierced edge not clean and smooth, i.e., any burr, drag, cutter, or file mark	-	*	-
	Design or shape is damaged, distorted or altered by piercing operation	x	-	-
	Forcer used when not specified or insignia forced in other than specified location	-	*	-
	Impression of the forcer (when applicable) is not symmetrical with the design on the front	-	-	x
	Metal torn or broken out by use of forcer	-	-	x
	Any component missing, bent, twisted, broken, or malformed	-	*	-
	Solder does not completely unite parts, i.e., any perceptible opening between component parts	-	*	-
	Area burned or reduced in soldering	-	*	-
	Soldered joint not clean or smooth	-	-	x
	Excess solder or excess flux not removed	-	-	x
BACK AND EDGES:				
Finish	Not clean, i.e., discoloration, such as spot or stain	-	-	x
	Not finished as specified	x	-	-
Attaching device	Defective, i.e., any part damaged or malformed which affects use or which may be expected to affect use	x	-	-
	Deformed, i.e., any part misshaped, damaged, bent, but not affecting use	-	-	x
	Not type specified	x	-	-
	Not positioned as specified	x	-	-
	Not attached in specified manner, i.e., not soldered or swaged as required	x	-	-

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TABLE III. DEFECTS (Continued)

EXAMINE	DEFECT	CLASSIFICATION Major	(*)	Minor
FRONT:				
	Metal strip not clinched around shank of prong (when required).....	x	-	-
	Clutch fastener will not engage or is loose fit on prong.....	x	-	-
	Clutch pierced through by prong.....	-	-	x
	Swaged prong can be removed by hand.....	x	-	-
	Both swaged prongs are loose and wobble.....	-	-	x
	Threads on post or nut are stripped, crossed, damaged or not specified size or type.....	x	-	-
	Any component omitted, misplaced, operation omitted or not properly performed.....	-	*	-
Identification Marking	Missing, incorrect, illegible, misspelled, not accomplished as specified, or not placed as required.....	-	-	x

4.5.2.1.2 Examination of insignia for defects in dimensions. Any dimension which is not within the specified tolerance shall be classified as a defect.

4.5.2.2 End item testing. Testing of the completely fabricated insignia shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit for the test for hard soldered or fused joints shall be one unit. All items shall be tested and one defect will be scored regardless of how many prongs on that item failed. The sample unit for all other applicable end item tests shall be four. For other than the test for fused prongs, the requirements shall be applicable to the individual unit.

TABLE IV. TESTING OF END ITEM

Characteristic	Requirement Paragraph	Test Method	# determ Per Sample /Unit	Results Reported as
Test for joints:				
Soldered joints	3.5.3	4.6.4.1	1	Pass or fail
Electronically fused joints	3.5.3	4.6.5	1	Pass or fail
Test for attaching devices:				
Prong and clutch	3.5.5.4	4.6.9.1	1	Pass or fail
Integral prong	3.5.5.2	4.6.3.1	8 1/	Pass or fail
Test for screw posts and thumb nuts	3.5.5.3	4.6.9.2	1	Pass of fail

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TABLE IV. TESTING OF END ITEM (Continued)

Characteristic	Requirement Paragraph	Test Method	# determ Per Sample /Unit	Results Reported as
Test for baked enamel	3.6.1.4	4.6.8	1	Pass or fail
Test for epoxy resin	3.6.1.3	4.6.6	1	Pass or fail
Silver plating thickness	3.6.2.5	STD comm'l	1 2/	Pass or fail
Nickel plating thickness	3.6.2.3	STD comm'l	1 3/	Pass or fail
Test for nickel underplating	3.6.2.4	STD comm'l	1	Pass or fail
Tests for lacquer:				
Liver of sulfur	3.6.3	4.6.10.1	1	Pass or fail
Tackiness test	3.6.3	4.6.10.2	1	Pass of fail
Test for fused joints	3.5.3	4.6.5	1	Pass or fail
Acid test:				
Gold plating, 24 Karat	3.6.2.1	4.6.2	1	Pass or fail
Gold plating, 22 Karat	3.6.2.2	4.6.2	1	Pass or fail

1/ Insignia are to be tested per sample unit as two prongs per insignia.
2/ In the event of dispute, the test method shall be Test Method 521.1 of FED-STD-151.
3/ In the event of dispute, the test shall be conducted in accordance with ASTM-B-487.

4.5.3 Examination for count of insignia for intermediate containers. Paired insignia and insignia of one design and finish packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (intermediate containers). Any box containing less than the specified or marked quantity of insignia or pair of insignia mismatched where applicable, shall be classified as a defect. The lot size shall be the number of intermediate containers.

4.5.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

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EXAMINE	DEFECT
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of envelops or container flaps, insecure seating of fastening device, tissue wrapping incomplete, voids inadequately filled, bulging or distortion of containers, loose strapping or taping, inadequate strapping.
Contents	Number of intermediate packages is more or less than required.
Weight	Net weight exceeds requirements.

4.5.5 Palletization examination. An examination shall be made to determine that the palletization is in compliance with the Section 5 requirements. Defects shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot.

EXAMINE	DEFECT
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.6 Tests.

4.6.1 Chemical composition test for copper base alloy. Chemical composition shall be determined in accordance with Test Method 111 of FED-STD-151. Results shall be evaluated to determine compliance with the requirements specified in 3.4.2.

4.6.2 Acid test for gold plating. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch (0.16 cm) in diameter on three (3) different spots on the plated surface allowing the drops to remain for not less than one (1) minute during which time the surface of the item shall be inspected to determine compliance with the requirements in 3.6.2.2 and 3.6.2.3. A minimum of two spots must withstand the acid test. The test acid shall be applied at room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius) and shall consist of a solution containing 50 percent by volume of chemically pure nitric acid (specific gravity 1.42) and an equal volume of distilled water. If the test insignia has been lacquered, the lacquer shall be removed prior to testing.

4.6.3 Test of Die Casting.

4.6.3.1 Test for integral prongs. Insignia to be tested shall be placed in a jig which allows no movement of the insignia and position prongs at a 30 degree angle

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above the horizontal. A vertical 10 pound load shall be applied to the top third of each prong. The prongs shall be examined to determine compliance with 3.5.5.2.

4.6.3.2 Direct micro thickness test for copper plating. Copper plating thickness shall be determined in accordance with Method 521 of FED-STD-151. Results shall be evaluated to determine compliance with 3.5.2.

4.6.4 Test for soldered joints.

4.6.4.1 Hard solder. Hard soldered items shall be placed in an oven maintained at 1075 degrees Fahrenheit ± 5 degrees Fahrenheit (561.7 degrees Celsius ± 2.78 degrees Celsius) for 15 minutes. While at this temperature the item shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.5.3.

4.6.4.2 Soft solder. Soft soldered items shall be placed in an oven maintained at 365 degree Fahrenheit ± 5 degrees Fahrenheit (184.98 degrees Celsius ± 2.78 degrees Celsius) for 15 minutes. While at this temperature the item shall be lifted by the superimposed design and shall be inspected to determine compliance with the requirements specified in 3.5.3.

4.6.5 Test for electronically fused joints. Item to be tested shall be anchored on a horizontal surface. All prongs on each end item shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45 degrees to each side of the vertical) until the prong breaks. The fused joint shall then be examined to determine compliance with 3.5.3. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.6.6 Test for shore hardness of epoxy resin. Shore hardness shall be determined in accordance with ASTM-D-2240. Results shall be evaluated to determine compliance with the requirements of 3.6.1.3.

4.6.7 Acetone test. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton and enamel shall then be examined to determine compliance with 3.6.1.4.

4.6.8 Test for baked enamel. Draw a knife edge across the face side of the specimen making four parallel scores 1/16 inch (0.16 cm) apart. Repeat this process drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.6.1.4.

4.6.9 Test for attaching device.

4.6.9.1 Test for prong and clutch type. The clutches shall be removed and replaced ten times from the prong using the clutch release wings. An attempt shall then be made to remove the clutch by hand without first depressing the release wings. An inspection shall be made at this time of the clutch and prong to determine compliance with the requirements specified in 3.5.5 or 3.5.5.4.

4.6.9.2 Test for screw post type and thumb nut. The thumb nut shall be assembled on the screw post, during which the operation of screw threads shall be observed to determine compliance with the requirements specified in 3.5.5.3.

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4.6.10 Test for lacquer.

4.6.10.1 Liver of sulfur test. Items to be tested shall be immersed in a 2 percent by weight chemically pure liver of sulfur (potassium sulfide) water solution at a temperature of 100 degrees Fahrenheit ± 10 degrees Fahrenheit (37.8 degrees Celsius ± 12.2 degrees Celsius) for a period of 3 minutes. The item shall then be removed and rinsed in warm, then cold, then hot water. The item shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture, allowed to dry at room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius) for one (1) hour and then be examined to determined compliance with 3.6.3.

4.6.10.2 Tackiness test. At room temperature (60 to 80 degrees Fahrenheit/15.6 to 26.7 degrees Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.6.3.

5. PACKAGING

5.1 Preservation. Preservation shall be Level C or Commercial, as specified (see 6.2).

5.1.1 Level C.

5.1.1.1 Unit packaging.

5.1.1.1.1 Army branch of service insignia. Twenty (20) Army branch of service insignia of one type only shall be mounted on a white sulfite board card, 6-1/2 inches wide by 5 inches (16.51 cm by 12.7 cm) and not less than 0.011 inch (0.028 cm) thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall be a minimum of 0.002 inch (0.005 cm) thick. All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

5.1.1.1.2 Insignia, service cap, Army; insignia, branch of service, insignia, service cap, Air Force; insignia, branch of service, collar and insignia, service cap, Marine Corps; and insignia, hat, Female, Army. One Army, Air Force or Marine Corps service cap insignia (one left hand and one right hand) or two Air Force branch of service insignia shall be mounted on a white sulfite board 2-1/2 inches (6.35 cm) square and not less than 0.011 inch (0.028 cm) thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall be a minimum of 0.002 inch (0.005 cm) thick. All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

5.1.1.1.3 Insignia, garrison cap, Marine Corps. One Marine Corps garrison cap insignia (left hand only) shall be mounted on a white sulfite board card, 2 inches (2.54 cm) square and not less than 0.011 inch (0.028 cm) thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall have a minimum thickness of 0.002 inch (0.005 cm). All seams and closures shall be heat sealed. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

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5.1.1.2 Intermediate packaging. Two hundred (200) Army branch of service insignia, fifty pair (50) Air Force branch of service insignia, one hundred (100) Female, Army hat insignia, forty (40) Marine Corps service cap insignia, fifty (50) Marine Corps collar branch of service or one hundred (100) Marine Corps garrison cap insignia, of one type only, shall be packaged in a setup paperboard box conforming to Variety 1 Style XDA of PPP-B-566. Box closure shall be in accordance with the appendix of the container specification. As an alternate method, setup boxes may be securely closed with strips of 1/2 inch (12.7 cm) minimum width pressure-sensitive, filament reinforced tape conforming to Type II, Class B of PPP-T-97. A minimum of two strips of pressure-sensitive tape shall be used. Each strip shall extend not less than 2 inches (5.08 cm) onto each box panel adjacent to the seams of the box.

5.1.2 Commercial. Insignia shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be Level C or Commercial as specified (see 6.2).

5.2.1 Level C. Insignia preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to Class domestic, Grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed the size and weight limitations of the container specification.

5.2.2 Commercial. Insignia shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.3 Palletization. Unitized loads shall be palletized on a 4-way entry pallet in accordance with load Type 1A of MIL-STD-147. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking for cleaning of gold plated items. To clean, rub gently with a soft, dry cloth. Do not clean with a chemically treated cloth or abrasive.

5.5 Marking. In addition to any special marking required by the contract or purchase order, unit packages, intermediate boxes, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM-D-3951 as applicable. Unit package marking may be printed on the reverse of the mounting card provided the printing is continuous and is legible through the bag.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Insignia covered by this specification are intended to be worn by enlisted personnel of the Armed Forces to indicate the wearer is a member of a particular service, corps or branch.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. The number and date of the applicable specification sheet (see 1.2 and 2.1).

d. When a first article is not required (see 3.2).

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

f. When palletization is required (see 5.3).

6.2.1 Contract data requirements. Data conforming to Department of Defense Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurement, the cognizant procuring activity will provide comparable data item descriptions. When so required, such data will be specified for delivery on a DD-1423 included in the contract.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.7.

6.5 Subject term (key word) listing.

Integral prong
Registering prong

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - IH
Air Force - 11
Navy - NU
Coast Guard - CG

Review activities:
Air Force - 32, 82, 99
Navy - MC
DLA - CT

Preparing activity:
Army - IH

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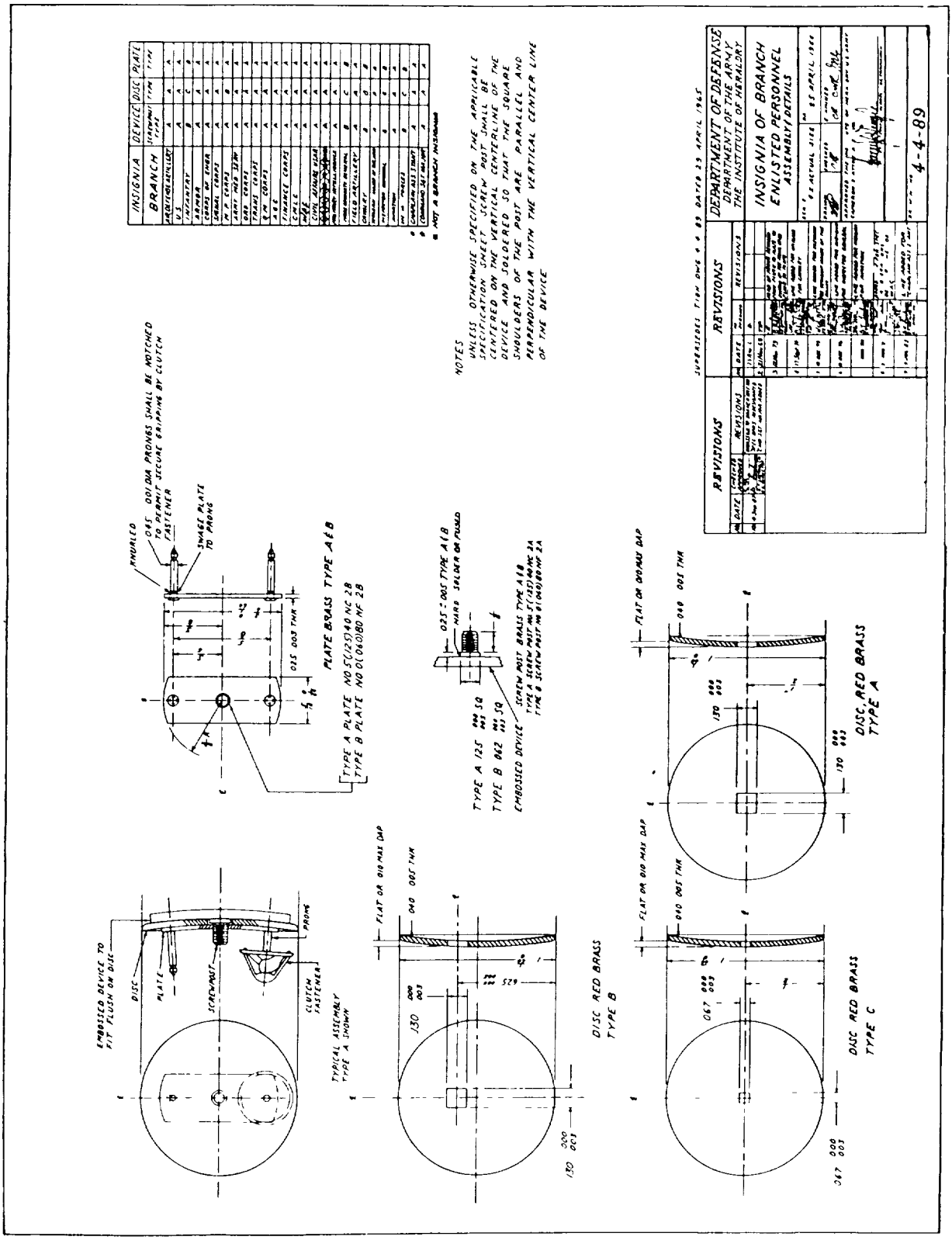


FIGURE 1. Pendant Ribbon Bars, Regular & Miniature, Assembly and Details

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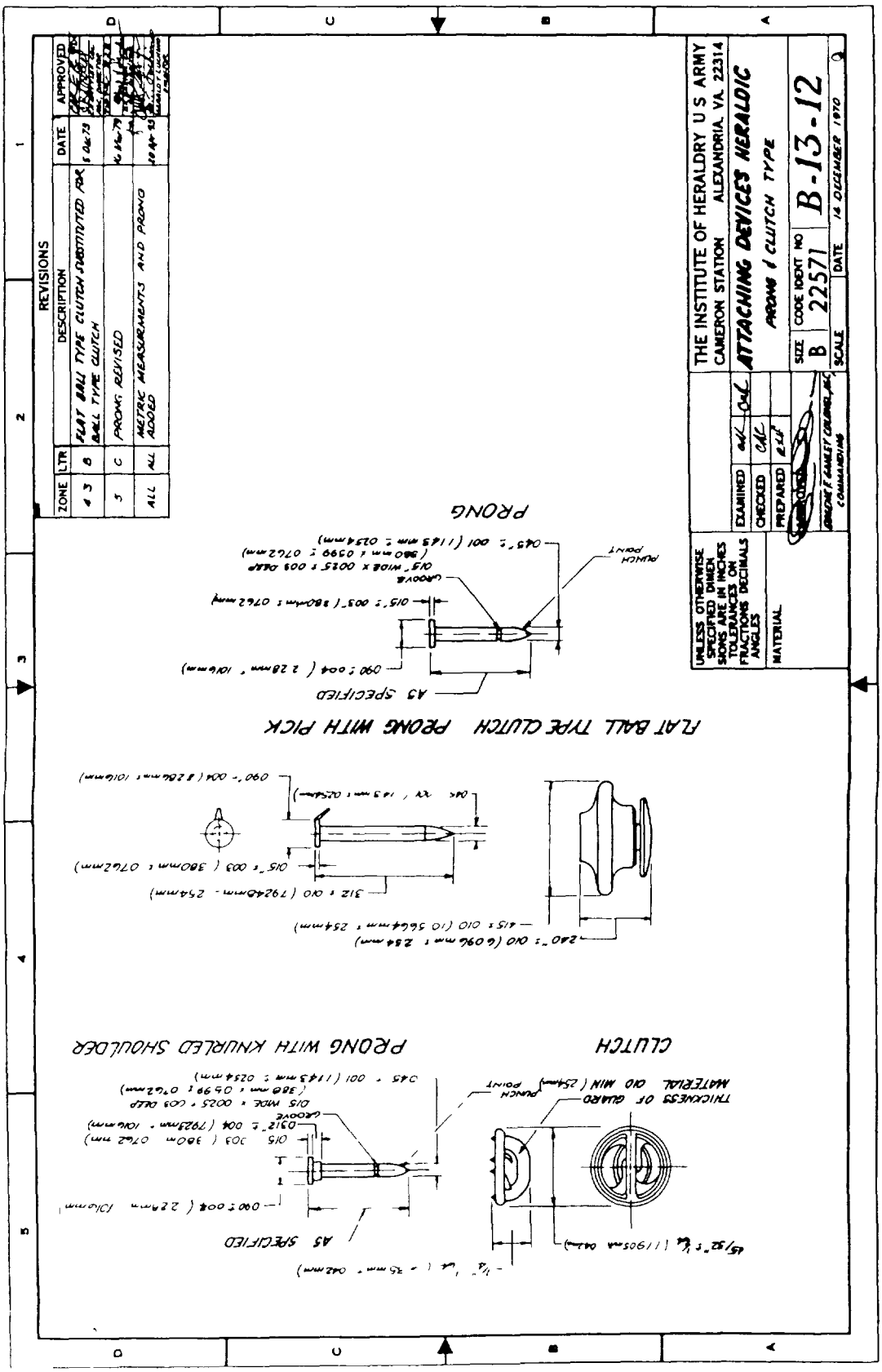


FIGURE 2. Attaching Devices, Heraldic, Prong & Clutch Type

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-I-15665H	2 DOCUMENT DATE (YYMMDD) 940429

3 DOCUMENT TITLE INSIGNIA, BRANCH OF SERVICE, HAT, GARRISON CAP, AND SERVICE CAP; ENLISTED PERSONNEL, GENERAL SPECIFICATION FOR

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
The Institute of Heraldry, U.S. Army	(703) 806-4990	DSN 656-4990
c ADDRESS (Include Zip Code) Technical & Production Division 9325 Gunston Road, Room S112 Fort Belvoir, VA 22060-5579	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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